

Improving Trial Recruitment & Engagement Through Operant Conditioning

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The Problem

- Recruitment sites must engage in complex learned behaviors to navigate RCT protocols.
- All sites have multiple competing interests for time and attention.
- Low reward structure for most stakeholders.
 - May seek to avoid unwanted attention
 - Some participation may be enough to trigger the reward structure with little extra reward for high performance (e.g., most sites only recruit a few patients)
 - Local contingencies more potent than remote ones

A Class of Solutions

- Operant conditioning combined with applied behavioral analysis can provide a straightforward system for enhancing desired behaviors, while extinguishing unwanted ones
- Applicable EVERYWHERE
 - Key to all human (machine) learning
- Although the principles are easy, the application can be challenging and requires thoughtful, systemized, **programs** that consider:
 - Target behaviors
 - Tailoring rewards
 - Reward schedules
 - Shaping
 - Maintenance

Actions Have Consequences: 'Contingencies'

Negative Punishment

Remove a desirable stimulus to decrease behavior.

Example: Take away attention at meeting.



Positive Punishment

Add an aversive stimulus to decrease behavior.

Example: Verbal reprimand

Decrease
behavior



Negative Reinforcement

Remove an aversive stimulus to increase behavior.

Example: Remove reporting duties after meeting goals.



Positive Reinforcement

Add a desirable stimulus to increase behavior.

Example: Praise team member after successful enrollment.

Increase
behavior

- (Remove)

+ (Add)

Target Behavior(s)

- A **target behavior** is a specific, observable action that you want to increase (or decrease) through reinforcement.
- In the context of clinical trial recruitment and engagement, target behavior(s) are the critical activities that directly contribute to trial success.
 - Clearly defined
 - Reliably measured
 - Reinforced (reinforceable) when they occur
- Examples
 - Screening Initiation
 - Enrollment Completion
 - Timely Data Entry
 - Follow-Up Compliance
- Each program should clearly identify target behaviors!

Identifying Reinforcement Strategies

Positive Reinforcers (Adding Desirable Outcomes):

- Monetary incentives (e.g., bonuses, gift cards)
- Public recognition and praise
- Career-related rewards (e.g., authorship, conference attendance)
- Team-based rewards (e.g., celebrations, meals, additional resources)
- Access to preferred tasks or opportunities (e.g., involvement in protocol development)

Negative Reinforcers (Removing Undesirable Conditions):

- Reducing burdensome administrative tasks
- Streamlining reporting requirements
- Providing schedule flexibility after meeting targets
- Minimizing oversight/supervision when performance is high

Avoid punishment if possible!

Reinforcement Schedules

- Once target behaviors and reinforcers are identified, the next critical step is to select an appropriate reinforcement schedule. The schedule determines how often and under what conditions reinforcement is delivered.
- **Fixed Ratio (FR)**
 - Reinforcement is delivered after a set number of responses (e.g., 1:1 ratio — every behavior is reinforced).
 - **Strengths:** Rapid acquisition of behavior; clear contingency between action and reward.
 - **Weaknesses:** Risk of dependence on reinforcement; behavior may drop if reinforcement stops (extinction).
- **Variable Ratio (VR)**
 - Reinforcement is delivered after a varying and unpredictable number of responses.
 - **Strengths:** Produces high and steady response rates; resistant to extinction.
 - **Weaknesses:** Less effective for initial learning; may frustrate some team members early on.
- **Fixed Interval (FI)**
 - Reinforcement is available after a fixed amount of time has passed since the last reinforcement (e.g., weekly bonuses if targets are met).
 - **Strengths:** Easier to implement; builds consistency.
 - **Weaknesses:** Responses often increase as the interval ends (scalping effect); low rates of behavior right after reinforcement.
- **Variable Interval (VI)**
 - Reinforcement is available after varying, unpredictable intervals of time.
 - **Strengths:** Steady and moderate rates of responding; resistant to extinction.
 - **Weaknesses:** Less control over when behaviors occur; may not produce rapid increases in behavior.

Summary of Strategy:

- Use **Fixed Ratio (FR)** schedules early in the program to shape and rapidly establish target behaviors.
- Transition to **Variable Ratio (VR)** schedules to sustain behaviors and prevent extinction over time.
- Consider **Interval Schedules** for ongoing maintenance tasks that benefit from steady engagement.

Introducing a Reinforcement Program

- To systematically apply operant conditioning principles to enhance recruitment and engagement, it is valuable to formalize these strategies into a structured reinforcement program.
 - Such a program ensures consistency, transparency, and scalability across recruitment sites.
- The core elements of the program involve:
 - **Selecting Target Behaviors:** Define the specific recruitment and engagement behaviors that are critical to trial success. The selected behaviors should be observable, measurable, and directly linked to site performance goals.
 - **Identifying Reinforcers:** Determine the rewards that are most meaningful to the individuals responsible for these behaviors. Different team members may respond to different reinforcers (e.g., monetary bonuses for coordinators, public recognition or authorship opportunities for investigators). Matching reinforcers to individual motivations increases the likelihood of success.
 - **Choosing a Reinforcement Schedule:** Establish how and when reinforcements will be delivered. Fixed ratio schedules (e.g., 1:1) may be used initially to rapidly build desired behaviors, while variable ratio schedules may later maintain and strengthen these behaviors over time.
 - **Operationalizing the Program:** Develop a clear implementation plan. This includes defining who will monitor performance, how reinforcement will be triggered and delivered, and how progress will be tracked and reviewed. Regular feedback loops and periodic adjustments will help ensure the program remains aligned with site needs and trial objectives.

Common Challenges & Opportunities

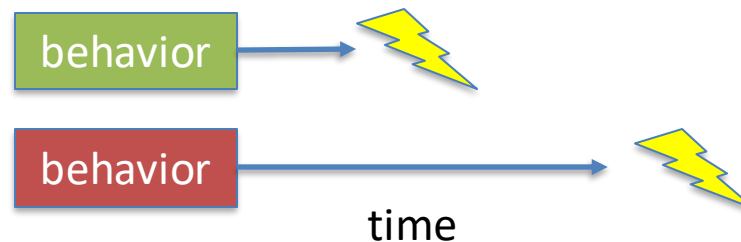
- The next few slides address issues that will arise during the implementation of an operant conditioning program
- This is the 'advanced stuff'

Tailoring Reward Strategy

- Different team members are motivated by different reinforcers
 - Too often, rewards are delivered only to the team leader, when they should be available for everyone who plays a role in recruitment
- Utility should be considered
 - Reinforcers are most reinforcing for those with the least amount of the reinforcer
 - Example: a \$5 gift card means the most to individuals with lower salaries

Immediacy of Reward Matters

- For reinforcement to be effective, **timing is critical**.
 - The closer the reward is delivered to the occurrence of the target behavior, the stronger its impact on learning and behavior repetition. When reinforcement is immediate, individuals can easily connect their action to the outcome, strengthening the behavior.
- A delay between the behavior and the reward diminishes the potency of the reinforcement.
 - Delayed rewards may weaken the association, reduce motivation, and make it harder for the desired behavior to become habitual, especially for new or less established behaviors.
- In clinical trial recruitment, reinforcing target behaviors such as rapid screening or prompt data entry should be done as soon as possible (e.g., same day recognition, instant feedback or rewards) to maximize effectiveness.



The Power of Attention as a Reinforcer

- In any reinforcement program, it is crucial to recognize that **attention itself is a powerful and often underestimated reinforcer**. Human behavior is naturally sensitive to social cues — praise, recognition, acknowledgment, and even subtle reactions from supervisors and peers can all influence behavior, often without formal reward systems.

Positive Attention as Reinforcement

- Public praise during meetings, acknowledgment in emails, or personal thanks from leadership can strongly reinforce target behaviors.
- These social reinforcers often feel more immediate and meaningful than monetary or material rewards, especially in team environments.
- When used intentionally, attention can efficiently shape and maintain desired behaviors.

Inadvertent Reinforcement of Non-Target Behaviors

- Attention is not always positive — but **any form of attention can reinforce behavior**, including unwanted behaviors.
- Complaints and excuses may be unintentionally reinforced if they reliably elicit problem-solving attention from trial leaders.
- Low-effort or inconsistent performance may persist if still met with attention (e.g., frequent check-ins, lenient oversight).

Key Consideration:

- Be mindful and deliberate about how attention is distributed.
- Provide **consistent positive attention to target behaviors**, while minimizing attention (or redirecting it constructively) when non-target or counterproductive behaviors occur.

Shaping Complex Behaviors: Building Toward Enrollment

- Enrollment in a clinical trial is not a single action, but rather a **complex chain of smaller, prerequisite behaviors** — each of which must occur reliably for the enrollment process to succeed. In behavioral terms, enrollment is a **composite behavior**, made up of multiple atomic steps that often require separate reinforcement and support.

What is Shaping?

- Shaping involves **reinforcing successive approximations** of a target behavior.
- Rather than waiting for the full behavior to occur for reinforcement, smaller steps along the way are identified and selectively reinforced.
- This gradually guides individuals or teams from initial, simpler behaviors toward more complex and integrated actions.

Example: Shaping Toward Successful Enrollment

- Step 1: **Identifying potential participants** — Encourage staff to flag and refer eligible patients regularly.
- Step 2: **Screening for eligibility** — Reinforce careful, accurate application of inclusion/exclusion criteria.
- Step 3: **Completing consent procedures** — Reinforce attempts at informed consent discussions.
- Step 4: **Executing enrollment and documentation** — Reinforce timely and complete data entry and confirmation of enrollment.

Why Shaping is Critical

- Without reinforcing each step, weak links in the behavioral chain (e.g., inconsistent screening or missed consent discussions) can derail the overall process.
- Shaping allows reinforcement to occur **early and often**, even before full enrollment occurs, keeping motivation high and providing continuous behavioral feedback.

Key Principle: When target behaviors are complex, focus on reinforcing the building blocks first. Shape each atomic behavior until reliable, then reinforce integration into the full, complex behavior.

Leveraging Social Learning in Operant Conditioning

- While operant conditioning typically focuses on **direct reinforcement of behavior**, powerful effects can also be achieved when reinforcement is delivered **in public view**, allowing others to observe and learn from these contingencies. This process, often referred to as **vicarious reinforcement**, draws upon principles of **social learning theory**.

Social Learning and Vicarious Reinforcement

- People learn not only through their own experiences, but also by **observing others**.
- When reinforcement is provided **visibly and socially** (e.g., public praise, recognition in meetings), others are likely to infer the value of the reinforced behavior.
- This can lead to an increase in the observed behavior across the network, even among those not directly reinforced.

Application to Clinical Trial Recruitment

- **Public Recognition:** Calling out high-performing sites during network-wide calls or newsletters highlights desirable behaviors (e.g., meeting enrollment targets, completing data entry on time).
- **Peer Influence:** Sites may emulate peers who are praised, especially if recognition is tied to valued outcomes like leadership visibility or future study opportunities.
- **Modeling Desired Behaviors:** Sharing best practices and case examples from reinforced sites provides clear behavioral models for others to follow.

Benefits of Social Delivery of Reinforcement

- **Amplifies reach:** One reinforcement episode can influence many observers.
- **Promotes norm formation:** Desired behaviors become part of “what sites like us do.”
- **Encourages healthy competition and motivation.**

Key Principle: Public reinforcement doesn't just shape individual behavior — it also sends powerful social signals that encourage widespread adoption of target behaviors across teams and sites.

Using Novelty and Surprise in Reward Selection

- As reinforcement schedules shift from frequent to more variable delivery, maintaining engagement becomes critical. One powerful method is to introduce **novel or surprising reinforcers within an existing reinforcement schedule**. This approach keeps the contingency predictable (i.e., behavior → reward will occur), while making the reward itself unpredictable and fresh.

Why Vary the Reward?

- **Prevents habituation:** Repeated use of the same reinforcer can lead to diminishing returns as individuals become accustomed to it.
- **Sustains motivation:** Unexpected or novel rewards renew interest and make reinforcement events more memorable and meaningful.
- **Enhances the impact of variable schedules:** When reinforcement delivery is already less frequent, varying the reward can make each event more salient.

Examples of Novelty within Reinforcement Schedules

- Maintain the **Variable Ratio** schedule, but occasionally substitute a different reinforcer (e.g., instead of the usual gift card, provide public recognition, swag, or a unique experience).
- For sites accustomed to routine incentives, **surprise upgrades** (e.g., doubling a bonus unexpectedly) can reinvigorate motivation.
- Introduce **thematic or time-limited rewards** (e.g., seasonal incentives, mystery rewards revealed upon achievement).

Key Principle: While reinforcement schedules control *when* rewards are delivered, novelty in the reward itself controls *how exciting or motivating* each reinforcement event will be. Using novel or surprising rewards within a stable schedule combines the benefits of consistency and engagement, supporting long-term adherence to target behaviors.

The Role (and Limitations) of Punishment

- While operant conditioning includes punishment as a core concept, defined as any consequence that decreases the likelihood of a behavior, its application in the context of clinical trial recruitment and engagement requires caution and thoughtful consideration.

Why Punishment is Generally Discouraged in Recruitment Programs

- **Undermines motivation:** Punitive measures may create fear, resentment, or disengagement, all of which can undermine team morale and intrinsic motivation.
- **Damages relationships:** Clinical research teams rely on collaboration and trust. Punishment can strain relationships between site staff and study leadership.
- **Fails to teach desired behavior:** Punishment stops unwanted behaviors, but does not guide staff toward what they should be doing instead.
- **Risk of withdrawal:** Unlike reinforcement, which encourages participation, punishment may lead to avoidance, withdrawal, or minimal compliance.

When (and How) Punishment May Be Appropriate

- Reserved for clear, serious breaches of protocol or ethical standards, not for performance variability in complex tasks like recruitment.
- Should be procedural and fair, not personal or punitive (e.g., retraining requirements rather than public reprimand).
- Should be accompanied by constructive feedback and reinforcement for alternative, desirable behaviors.

Key Principle: In the context of recruitment and engagement, punishment is unlikely to drive positive change. Instead, focus on reinforcement-based strategies to encourage desired behaviors and use punishment sparingly — only when necessary to uphold essential standards.

Implementing and Managing the Reinforcement Program

- While defining target behaviors and selecting reinforcers are foundational steps, the success of any operant conditioning program ultimately depends on how well it is operationalized and managed day-to-day. This requires clear procedures and assigned responsibilities.

Define Roles and Responsibilities

- Identify who will monitor behavior and determine when reinforcement should be delivered (e.g., trial manager, site liaison, DCC, or CCC).
- Explicitly designate individuals responsible for delivering reinforcement, this may include both local site leadership and central trial leadership. Tip: Different team members may bring novelty to the reinforcer!

Establish Reinforcement Procedures

- Define clear criteria for reinforcement to ensure fairness and transparency (e.g., timely enrollment, clean data entry, protocol adherence).
- Determine delivery methods (e.g., during site calls, via email recognition, automated bonuses).
- Develop a documentation and tracking process to ensure consistency and to allow for later evaluation.

Maintain Engagement Through Feedback Loops

- Regularly review performance metrics to ensure reinforcement efforts are aligned with trial goals.
- Provide ongoing feedback to sites, reinforcing positive trends and addressing challenges collaboratively.

Plan for Scalability

- Use standardized processes to make the reinforcement program scalable across multiple sites while allowing flexibility for site-specific adjustments.

Key Principle: A reinforcement program must be seen as credible, consistent, and fair. Clear management processes help ensure that reinforcement strengthens site engagement without introducing confusion or perceived favoritism.

Ethical Considerations in Using Reinforcement

- While operant reinforcement is ubiquitous, its use in clinical trial settings must be carefully managed to ensure that it is ethical, equitable, and supportive of scientific integrity.

Avoiding Coercion or Undue Influence

- Reinforcers should be meaningful but **proportionate** — overly large incentives may create pressure to cut corners or compromise protocol adherence.
- Reinforcement should reward **process behaviors** (e.g., completing data entry on time), not outcomes that may be influenced by participant factors (e.g., recruitment at all costs).

Ensuring Equity and Fairness

- Reinforcement opportunities should be available to all sites and roles, with adjustments to account for differences in site size, resources, and staffing.
- Reinforcement should reflect effort and process, not just raw volume, to avoid disadvantaging smaller or resource-limited sites.

Supporting Autonomy and Intrinsic Motivation

- External reinforcement should **complement (not replace)** intrinsic motivators such as pride in work, commitment to science, and concern for participant welfare.
- Reinforcement programs should **fade appropriately** as desired behaviors become habitual and self-sustaining. Culture is paramount.

Key Principle: Reinforcement should enhance — not distort — trial processes. Ethical reinforcement programs support a culture of excellence while respecting the professional integrity and autonomy of research staff.

Thank you!